

A coupled model of **Surge, **WA**ve and **T**ide (SuWAT)**
for storm surges and waves

User manual

Version 2031

updated : 2014.03.26
updated : 2014.04.02
updated : 2014.04.08
updated : 2014.07.12
updated : 2016.03.24
updated : 2016.04.07

SuWAT

programmed by
Sooyoul Kim

Tottori University, April 2008 to present
Assistant Professor

Disaster Prevention Research Institute, Kyoto University,
Assistant researcher, 2007 - 2008
PhD student, 2004 - 2007

sooyoul.kim@sse.tottori-u.ac.jp
klaatujk@gmail.com

Contents

1. Basic equations of SuWAT.....	5
2. Conditions of use	6
3. Installing and compile Instruction	7
3.1 MPICH	7
3.2 Compile instruction.....	7
4. Framework of SuWAT	8
4.1 Structure of files	8
4.2 Input files for surge module	9
4.2.1. domain_01_table.geo	10
4.2.2. domain_01_table.mp	10
4.2.3 domain_01_table.tf	11
4.2.4 domain_01_runup.table.bin.....	12
4.2.5 bottomcoef_01_table.dat	12
4.2.6 event_name.srt.....	12
4.2.7 case.event_name.srt.....	13
4.3 Input files for wave module	13
4.3.1 domain_01_table.geo.swan	13
4.3.2 initial files	14
4.3.3 path files.....	14
5. Control files	15
5.1 filename_d0#.con	15
5.2 ample_and_time	15
5.3 filename_wind_pr_d0#	17
5.4 #d.con	17
5.5 #d.nest	34
5.6 swaninit#.....	37
5.7 #d.SWN.....	38
5.7.1 The command for model input	40

5.7.2 The commands for output requests	41
6. Output files for the storm surge module.....	43
6.1 d0#.out.nest.swt	43
6.2 Spatial distributions I.....	43
6.3 Time series I.....	44
6.4 Spatial distributions II.....	44
6.5 Time series II.....	46
6.6 Summary of "event_name.srt"	47
7. Output files for the wave module.....	48
7.1 FORCE: wave-induced radiation stresses. No option.....	48
7.2 HSIGN: significant wave height (option)	48
7.3 PDIR: direction (option)	48
7.4 RTP: period (option).....	49
8. Frequent Q&A.....	50
8.1 Garden-Sprinkler-Effect (GSE)	50

1. Basic equations of SuWAT

Please refer to the following manual and selected papers:

1. Sooyoul Kim, Nobuhito Mori, Hajime Mase, Tomohiro Yasuda (2015) The role of sea surface drag in a coupled surge and wave model for Typhoon Haiyan 2013, *Ocean modelling*, 96, p.65-84, 10.1016/j.ocemod.2015.06.004
2. Sooyoul Kim, Y., Matsumi, T., Yasuda, H., Mase (2014), Storm surges along the Tottori coasts following a typhoon, *Ocean Engineering*, Vol. 91, 133-145, DOI: 10.1016/j.oceaneng.2014.09.005
3. Nobuhito Mori, Masaya Kato, **Sooyoul Kim**, Hajime Mase, Yoko Shibutani, Tetsuya Takemi, Kazuhisa Tsuboki, Tomohiro Yasuda (2014) Local amplification of storm surge by Super Typhoon Haiyan in Leyte Bay, *Geophysical Research Letters*, Vol. 41, 5106-5113, DOI: 10.1002/2014GL060689
4. Tomohiro Yasuda, Sota Nakajo, SooYoul Kim, Hajime Mase, Nobuhito Mori and Kevin Horsburgh (2014) Evaluation of Future Storm Surge Risk in East Asia based on State-of-the-art Climate Change Projection, *Coastal Engineering*, Vol. 83, 65-71
5. Kim, S.Y., Yasuda, T., Mase, H., (2010) Wave set-up in the storm surge along open coasts during Typhoon Anita, *Coastal Engineering*, Vol., 57, 631-642, 10.1016/j.coastaleng.2010.02.004
6. Kim, S.Y., Yasuda, T., Mase, H., (2008) Effects of Tidal Variations on Storm Surges and Waves, *Applied Ocean Research*, 30,311-322.
7. 金 洙列・安田誠宏・間瀬 肇 (2008) : 潮汐・高潮・波浪結合モデルによる土佐湾異常高潮の追算, 海岸工学論文集, 第55巻, pp.321-325.
8. 金 洙列・高山知司・安田誠宏・間瀬 肇 (2007) : 高潮と波浪に及ぼす大潮汐変動の影響に関する研究, 海岸工学論文集, 第54巻, pp.276-280.
9. Dissertation, Effect of large tidal variation on the storm surge in the western coastal sea of Korea, Kyoto University, 2007.
10. User manual of SWAN developed by Delft University.
11. User manual of tsunami / storm surge model developed by Toukai University.

2. Conditions of use

- The tidal prediction (the present version supports the Global area and the East Asia area).
- The storm surge and wave simulations with or without the tide.
- The empirical drag coefficient estimated by Honda-Mitsuyasu formula and the wave dependent drag coefficient (Janssen,1999).
- Inclusion of wave-induced radiation stress in the storm surge model.
- The parallelized and nested scheme employed by MPI (message passing interface).
- The developer takes no responsibility of damage or loss by the use of this program.

3. Installing and compile Instruction

3.1 MPICH

Before the SuWAT model is compiled, users must install MPICH1 or MPICH2 on the system.

Downloads

MPICH1 (<http://www-unix.mcs.anl.gov/mpi/mpich1/mpich-nt/>)

MPICH2 (<http://www-unix.mcs.anl.gov/mpi/mpich2/>).

3.2 Compile instruction

SuWAT is supported on Windows and Linux OS.

Refer to Makefile in the repository of SuWAT on linux.

Refer to the previous manual for Windows. The binary file for SuWAT is provided.

Contact to sooyoul.kim@sse.tottori-u.ac.jp to get the source code of SuWAT.

4. Framework of SuWAT

4.1 Structure of files

All folders are removed in the present version.

In addition, all files should be located under the working folder.

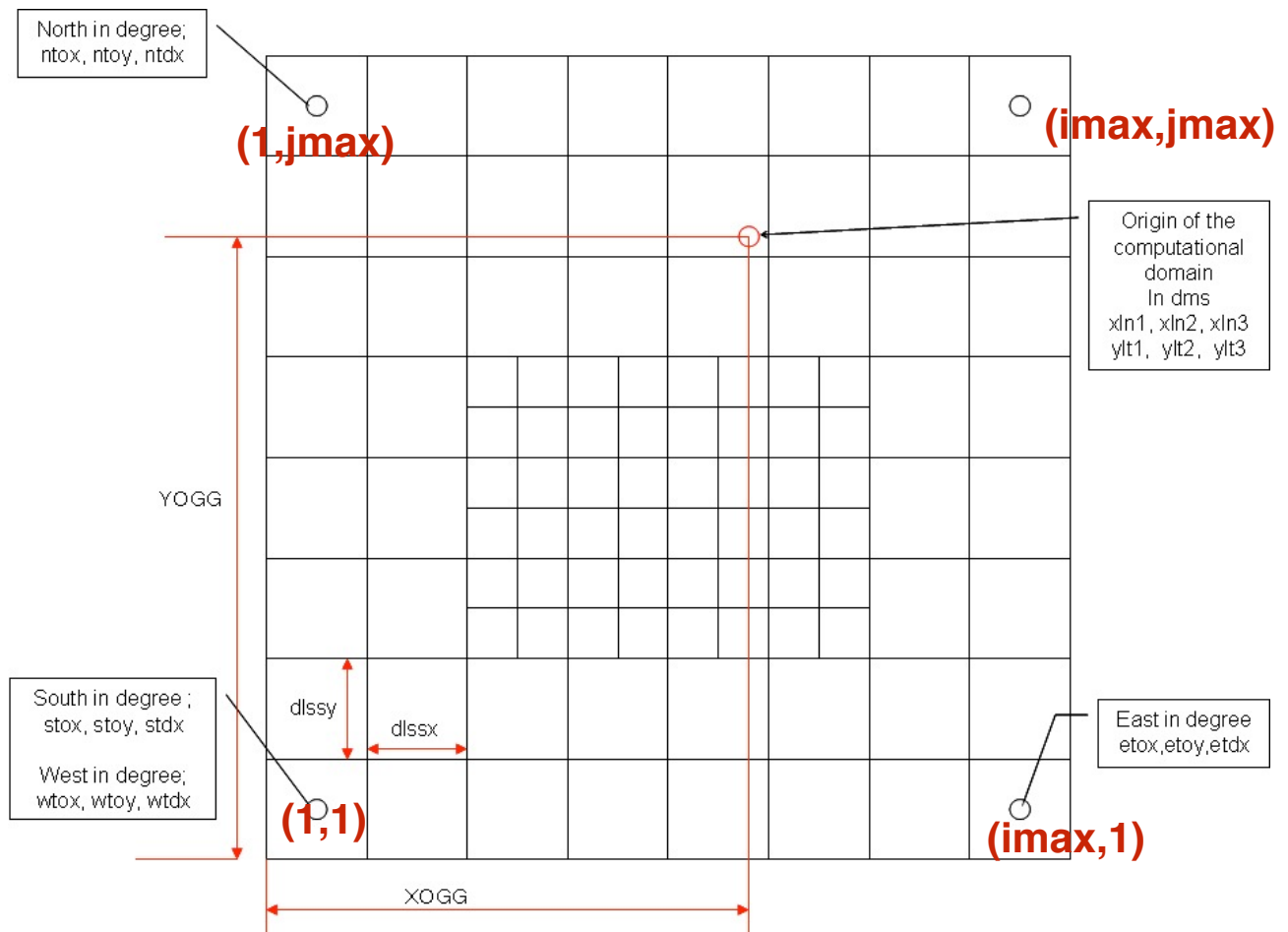


Fig. 4.1 Scheme of the computational grid in the storm surge module

4.2 Input files for surge module

The manual describes the flags for the surge and wave modules and the input and output files to simulate the storm surge and the wave. The space in the coupled model is discretized in Cartesian coordinates. The different grid sizes in x and y are available in the domain.

The data set of topography should be prepared and the following filenames are recommended:

	File name	Description
	domain_01_table.geo	water depth levels (The water depth is positive but the land height is negative).
	domain_01_table.mp	The masks for the land, the sea and the breakwaters. The mask is "0" for the land, "1" for the sea and "11 or 21 or 31" for the breakwater. See Fig. 4.2.
	domain_01_table.tf	The format is same with "domain_01_table.map" without headers but is used to print the spatial distribution of water surface levels.
	domain_01_runup.table.bin	The height of breakwaters (The height in centimetres above the sea surface level is positive but negative below it). No headers
	bottomcoef_01_table.dat	The spatial distribution of manning coefficients (the format is same with "domain_01_runup.table.bin") No headers
	event_name.srt	Typhoon information
	case.event_name.srt	The start time and the end time for the script of "controlfile4swan_#D.sh"

4.2.1. domain_01_table.geo

Two headers are needed. The water depth data is arranged at the next line. The water depth in meters is positive, but the land is negative.

```
IGD, JGD
XOGD, YOGD, DLSDX, DLSDY
314.450 322.340 332.060 345.970 359.880 341.560 366.640 ...
301.300 306.490 316.760 322.170 323.170 322.390 313.680 ...
301.000 308.730 305.110 308.750 312.190 297.470 286.590 ...
305.750 302.620 294.640 289.500 280.710 274.550 274.410 ...
227.680 263.680 260.630 253.760 260.510 263.380 264.680 ...
...    ...    ...    ...    ...    ...    ...    ...
...    ...    ...    ...    ...    ...    ...    ...
```

	Variable	Description
	IGD	The number of grids in row.
	JGD	The number of grids in column.
	XOGD	The horizontal length in meters from the origin of the computational grid to the left coner of grid (1,1) in Fig.4.1.
	YOGD	The vertical length in meters from the origin of the computational grid to the left coner of grid (1,1) in Fig.4.1.
	DLSDX	The grid size along the horizontal direction
	DLSDY	The grid size along the vertical direction

4.2.2. domain_01_table.mp

The format is 'i2' with no space between two marks. There are two headers and then, the map data follows. In general, the sea grid should be 1, but the land grid is 0. In case of inundation calculation, the inundation regions may be 1. The breakwater is represented by 11, 21 and 31, respectively.

```
Example:
IGM, JGM
XOGM, YOGM DLSMX, DLSMY
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 2 1 2 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 3 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
.....    ...    ...    ...    ...
```

	Variable	Description
	IGM	The number of grids in row.
	JGM	The number of grids in column.
	XOGM	The horizontal length in meters from the origin of the computational grid to the left coner of grid (1,1) in Fig.4.1.
	YOGM	The vertical length in meters from the origin of the computational grid to the left coner of grid (1,1) in Fig.4.1.
	DLSDMX	The grid size along the horizontal direction.
	DLSDMY	The grid size along the vertical direction.

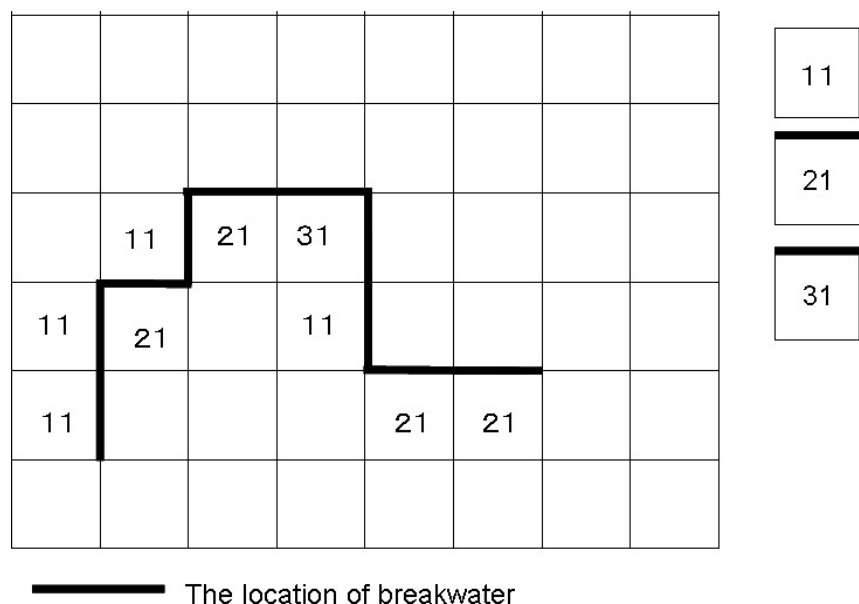


Fig. 4.2 Scheme of breakwater on the grid of "domain_01_table.mp"

4.2.3 domain_01_table.tf

In the case of wet and dry or tidal flat, the file determines the printout of water surface levels by the mask in this file. The file format is same with "domain_01_table.mp" but no headers and breakwaters.

Example:

```

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 ...
1 1 1 1 1 1 1 1 1 1 1 2 1 2 1 1 0 0 0 0 0 0 ...

```

4.2.4 domain_01_runup.table.bin

In this file, the heights of breakwaters in centimetres should be set if it is used. If no breakwater in the grid, it should be zero. If the height is negative below zero water level, SuWAT understand it is submerged structure. But it is not recommended. The file is optional as if activating the flag "runnp_con=1" in "#d.con". There are two headers and then, the map data follows.

Example:

```
IGM, JGM
XOGM, YOGM DLSMX, DLSMY
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...
0 0 0 0 0 0 0 0 0.1 0.1 0.1 0 0 0 0 0 ...
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...
```

4.2.5 bottomcoef_01_table.dat

In case of use of spatial varying bottom coefficients (Manning coefficient), this file is needed. The format is same with "domain_01_runup.table.bin". The file is optional as if activating the flag "bottomf_con=1" in "#d.con".

4.2.6 event_name.srt

In case of use of internally integrated parametric wind and pressure models, the typhoon information that is date, longitude, latitude, the difference between 1013 hPa (or 1014 hPa) and the central pressure (=1013 - Pc), the radius from the maximum wind to the centre, the moving speed of typhoon, is needed. Format is '(4I2, 2X, 2F10.3, 3F10.1)

Date (yymmddhhmm)	Longitude (degree)	Latitude (degree)	Depression rate of central pressure (hPs)	Distance to the station of the maximum wind speed from center (km)	Moving speed of typhoon (km/h)
0209240900	132.283	20.800	70.0	40.0	22.0
0209240901	139.200	22.800	70.0	40.0	25.0
0209240902	137.200	22.200	70.0	40.0	28.0
0209240903	140.105	40.100	60.0	50.0	30.0
...	...	...	...	...	...

Example:

3 ; the number of line

```

5909250859 132.283 14.071 10.0 75.0 73.0
5909250900 132.283 14.071 10.0 75.0 73.0
5909251000 132.474 14.699 12.0 75.0 73.0

```

4.2.7 case.event_name.srt

The file include the information of the start time and the end time after the spinup. No headers are needed. The file is read by the script "[controlfiles4swan_#D_ver2.sh](#)" to write out the SWAN input files.

Example:

```

195909250900 132.283 14.071 10.0 75.0 73.0
195909251000 132.474 14.699 12.0 75.0 73.0

```

4.3 Input files for wave module

	File name	Description
	domain_01_table.geo.swan	water depth levels (The water depth is positive but the land height is negative).
	input.cur	initial values of Zero for currents
	input.win	initial values of Zero for winds
	input.wle	initial values of Zero for water surface elevation
	seri.cur	path for input.cur
	seri.win	path for input.win
	seri.wle	path for input.wle

4.3.1 domain_01_table.geo.swan

Free format is used. No headers are needed. The water depth in meters is positive, but the land is negative. It may be needed to identify both of computational grids and non computational grids if necessary. When identifying both, the exception value is used as

"-99" in the region of the land. One dimensional channel is, however, ignored, if the exception value is used in SWAN. In order to avoid it, the water depth along the channel should have more than two grids along the channel width, which is sufficiently the high resolution grid, in the vicinity of the shoreline. However, it is recommended to no use the exception value to avoid the error in the computation. But the run speed may increase if the exception value is used. Simply "domain_01_table.geo" is able to be used without the headers.

Example:

72.26	66.77	88.89	90.34	90.29	89.01	...	...
67.32	75.33	76.44	79.57	80.53	79.85	...	...
64.77	48.46	6.66	68.53	70.26	66.29	...	...
-294.20	-414.42	-381.69	-65.54	51.76	46.25	...	...
-99.00	-99.00	-99.00	-37.13	31.58	29.38	...	...
-99.00	-99.00	-99.00	-117.86	-19.40	-73.51	...	...
-99.00	-99.00	-99.00	-99.00	-99.00	-99.00	...	...
-99.00	-99.00	-99.00	-99.00	-99.00	-99.00	...	...
-99.00	-99.00	-99.00	-99.00	-99.00	-99.00	...	...
-99.00	-99.00	-99.00	-99.00	-99.00	-99.00	...	...

4.3.2 initial files

It includes the initial values for the current, wind and water level as 0. Note that does not revise them. The files are written out by the script "[controlfiles4swan_#D_ver2.sh](#)" as running it.

```
input.cur
input.win
input.wle
```

4.3.3 path files

It includes the paths of input.cur, input.win and input.wle. Note that does not revise them.

```
seri.cur
seri.win
seri.wle
```

5. Control files

5.1 filename_d0#.con

In the "*filename#.con*", the "#" indicates the domain number. The following paths should be included in the file. (Note, the file order cannot be changeable)

Line No	File name
1	domain_01.table.tf
2	domain_01_table.geo
3	domain_01_table.mp
4	1d.con
5	1d.net
6	d01.out.nest.swt
7	bottomcoef_01_table.dat
8	ample_and_time
9	domain_01_table.bin

5.2 ample_and_time

The run length from the start and the end should be determined in the file. The end time should be same with the time in the "*event_name.srt*". All are the integer except that "*ampl*" is the real. If the external data of the wind and pressure is used, the start and end time should be in hours. The spinup run in SuWAT corresponds to the duration from "0" to "*start_hour* and *start_min*". During this run, the storm surge module only runs even though the run mode is the coupled run.

<i>ampl</i>	<i>passd</i>	not used but keep it
<i>start_hour</i> (in hour)	<i>start_min</i> (in minute)	
<i>center_hour</i>	<i>center_min</i>	not used but keep it
<i>end_hour</i> (in hour)	<i>end_min</i> (in minute)	

Example:

0.5 6
01 00
12 00
43 00



Fig. 4.3 Scheme of the spinup run in SuWAT (the spinup run corresponds to the duration from "0" to "start_hour and start_min").

If the tide is imposed on open boundaries, the tide calculation should be carried out 2~3 days from 0 to "start_hour" and "start_min" prior to the coupled or non-coupled runs of the storm surge and/or the wave desired by the user. This preceding run is known as "spinup run". Its duration may depend on the size of the computational domain. It is recommended that in case of the non-coupled run of the storm surge or the coupled run of surge and wave are also carried out with the spinup run even without the tide. It is recommended that all runs should have at least the 1 hour spinup run.

- ◆ Example: run with the tide
the spinup run length : 2 days
the surge run length : 1 day

- event_name.srt
3 : the number of line
5909250000 132.283 14.071 10.0 75.0 73.0
5909270000 132.283 14.071 10.0 75.0 73.0
5909280000 132.474 14.699 12.0 75.0 73.0
- ample_and_time
0.5 6
48 00
12 00
72 00

- ◆ Example : run without the tide
the spinup run length : 1 hours
the surge run length : 1 day

- event_name.srt
3 : the number of line
5909262300 132.283 14.071 10.0 75.0 73.0
5909270000 132.283 14.071 10.0 75.0 73.0
5909280000 132.474 14.699 12.0 75.0 73.0
- ampo_and_time#
0.5 6
01 00
12 00
25 00

5.3 filename_wind_pr_d0#

The paths of the external data for the wind and pressure should be assigned in these files. SuWAT basically searches "*filename_wind_pr_d01*", "*filename_wind_pr_d02*" and "*filename_wind_pr_d03*" (Note, three files are default). If more than three domains are used, the additional files can be prepared by the users. If you want, the external data can internally be interpolated within SuWAT. The order of the paths is not changeable.

- [filename_wind_pr_d01]
event_name.srt
external_data_name.u10
external_data_name.slp
- [filename_wind_pr_d02]
event_name.srt
external_data_name.u10
external_data_name.slp
- [filename_wind_pr_d03]
event_name.srt
external_data_name.u10
external_data_name.slp

5.4 #d.con

The calculation options for the storm surge and the wave :

- the tide,
- the storm surge using the empirical formula for the wind stress,
- the storm surge and the wave using the wave dependent drag coefficient (Janssen, 1999),
- the storm surge and the wave using the wave drag coefficient and wave-induced radiation stress,
- The options for wind and pressure data:

- parametric wind and pressure model integrated into SuWAT,
- external data files obtained from weather forecasting model such as WRF. The detailed description for the flag is given below:

d0#.con

Line number	Flags								
1	MDEL	LEVL	TIDE	INTAC	NESTIN G	TIDE_FIL E			
2	BS	SWANN T	HOT_S	CON_RS	CON_C D	CON_BE DCD	JAN_C ON	SWAN_LEV ELOFF	swan_idrag
3	OB_CO N2	OB_CON	TESTOB	TESTOB _WIND	FLATHE R_CON				
4	WIND_C ON	WIND_R EAD	WIND_W RITE	INTERTI ME	INPUTTI ME	PRES_C ON			
5	LEVELO FF	LIMIT_wi ndsp	c1	c2	rf_spuw	tydir			
6	OUTFFO RM								
7	ssl%tab	cur%tab	met%tab	swc%tab	ssl%max out	cur%ma xout	met%m axout		
8	rad%tab	psvssb%t ab	psvbot%t ab	flu%tab	sdc%tab	bdc%tab			
9	ssl%ts	cur%ts	met%ts						
10	rad%ts	psvssb%t s	psvbot%t s	flu%ts	sdc%ts	bdc%ts			
11	ch	manning _n	bottomf_ con	runup_c on					
12	IGR	JGR							
13	XOGG	YOGG	DLSSX	DLSSY					
14	DLT								
15	ITLM	PRCT							
16	IYEAR3	IMON3	IDAY3	IHOURL	HIMIN3				
17	ITS	ITSS							
18	WV								
19	XLN1								
20	YLT1								
21	ETOX	ETDY	ETDX						
22	STOX	STDY	STDX						

23	WTOX	WTOY	WTDX						
24	NTOX	NTOY	NTDX						
25	debugging								
26	idebug	jdebug	ktdebug						

LINE 1 MDEL LEVL TIDE INTAC NESTING TIDE_FILE SEDMODE

The flags are the integer.

MDEL 2: the storm surge
 1: the tsunami but not tested.

LEVL 1: the number of layers. Only one layer is used in the present version.

TIDE the flag determines the tide is calculated or not. It is also used if the user want only the meteorological data.
 0: the storm surge + the tide
 3: only the tide
 4: only the storm surge on the mean water level without the tide
 5: only the wind and sea level pressure without the surge computation.

INTAC With this flag the user indicates whether the effect of waves, such as the wave dependent drag coefficient and the wave-induced radiation stress, is considered in the storm surge and the wave calculation.
 0: with SWAN
 2: without SWAN.
 In case of 2, the Honda-Mitsuyasu formula is activated.

NESTING The flag indicates the present domain has the nested domain or not. As activating the flag $[NESTING] = 1$, the boundary values such as water levels and currents are printed out.
 1: the nested domain exists.
 0: the nested domain no exists.

TIDE_FILE As activating the flag, the user can designate the grid positions in degree along the open boundary.
 0: the grid positions in degree are linearly interpolated by the flags in the Line 21 ~ 24.

1: SuWAT reads a file of [tidal_boundary]. It has no format but its order must be subsequently east, south, west and north with two columns of [longitude latitude].

Example

145.4858 20.30898

145.4873 20.30987

SEDMODE This flag activates sediment transport modules.

0: not activated

1: activated

LINE 2 BS SWANNT HOT_S CON_RS CON_CD CON_BEDCD JAN_CON SWAN_LEVELOFF
SWAN_IDRAG

The flags are the integer.

BS the flag indicates that the present domain is imbedded on the coarse domain.

0: in case of the present domain is the outermost domain.

1: in other domains, 1 should be identified, except the first domain,

SWANNT the flag indicates the same function with [BS] for SWAN. In the case of which SWAN does not run using the open boundary value obtained from the coarse domain, [SWANNT] is 0 on the first domain. On the contrary, others should be identified 1.

HOT_S not used but do not omit. Hot start or cold start

0: not activate the hot start.

1: activate the hot start.

CON_RS with this flag the wave-induced radiation stresses in the storm surge module can be activated.

1: activate the wave-induced radiation stress in the storm surge module.

2: activate the wave-induced radiation stress in the storm surge module but the gradient of sea level pressures is not activated.

3: No activate both the wave-induced radiation stresses and the gradient of sea level pressures gradient in the storm surge module

0: No activate only the wave-induced radiation stresses.

CON_CD with this flag, the wave dependent drag coefficient (Janssen, 1999) can be activated.

0: the Honda-mitsuyasu formula is activated.

1: the wave dependent drag coefficient is activated. During the spinup, the honda-mitsuyasu formula works, even the flag is activated.

2: any drag coefficients are not activated in the storm surge module. It means that the wind stress is not included in the storm surge module.

CON_BEDCD with the flag, the bottom drag coefficient considering the wave-current interaction in the bottom can be activated in case of coupling runs of surges and waves. But still developing so set [CON_BED-CD=0]

JAN_CON with the flag, the WAM's wave theory can be activated in deep sea. It is recommended that the flag=0 in the operational run.
0: the SWAN's default theory is activated in deep sea.
1: the WAM's theory is activated in deep sea.

SWAN_LEVELOFF With the flag, the limitation of wind speeds can be activated. As activating the flag =1, the limitation work in the both modules of surges and waves. This flag only work in the coupling run.
0: no activate the limitation.
[wind speed in m/s]: activate the limitation. The user have to designates the specific wind speed. 30~35m/s are recommended.

SWAN_IDRAG with the flag, the equations of Wu or Zijlema can be selected. It is only activated in the coupling mode with SWAN.
0: Wu's equation
1: Zijlema et al 's equation

LINE 3 OB_CON2 OB_CON TESTOB TESTOB_WIND FLATHER_CON

The flags are the integer.

OB_CON2 with this flag, the users select a type of radiation condition methods on the open boundaries. It is recommended to use the Flather's radiation condition.
1: activate the Flather's radiation condition (1994) which uses both water levels and currents.
0: implies that water levels is only imposed between the coarse grid and the fine grid.

OB_CON	the users select the components of currents and water levels or only water levels to interpolate them from the coarse grid to the fine grid. It is recommended to use OB_CON=1 with currents and water levels. 1: the current and sea surface level 0: the sea surface level
TESTOB	with this flag, the users activate the print out of boundary values to the text file for the confirmation. 1: activate the print out of the text file 0 : no activate it.
TESTOB_WIND	with this flag, the users activate the print out of wind speeds to the text file for the confirmation. 1: activate the print out of the text file 0 : no activate it.
FLATHER_CON	the flag activates the equation along the open boundary among the original Flather, the one but the different scheme and the new equation by Kim et al. (2011). 3: activate the original equation by Flather (1975) 2: activate the original equation by Flather (1975) but the different grid scheme. 1: activate the new equation by Kim et al. (2011) 1: activate the OB for OTPS

LINE 4 WIND_CON WIND_READ WIND_WRITE INTERTIME INPUTTIME PRES_CON

The flags are the integer.

WIND_CON	with this flag, the user select the external data file achieved from a weather model or a parametric wind model internally integrated into SuWAT. 0: activate the parametric wind and pressure model by Mitsuyasu and Fujii's formula known as SGW. 1: activate the external data file. 2: activate Fujita's formula. 3: activate Mayers' formula.
WIND_READ	with this flag, SuWAT reads the external data files. 1: activate reading the files.

0: no activate reading the files.

WIND_WRITE with this flag, the users make SuWAT interpolate the wind and pressure data to the following nested domain.
1: activate the interpolation.
0: No activate the interpolation.

INTERTIME with this flag, the users can assign the interval time in minutes of the interpolation in case of use of the external data file.
For example, the interval of WRF data in the external file is 1 hour (*INPUTTIME*=60) in the present domain. However, the user tends to interpolate the data in 5 minute into the present domain. In this case, the flag should be [*INTERTIME* = 5].

INPUTTIME with this flag, the users should assign the Interval time in the external data file in minutes.

PRES_CON with this flag, the users can select the parametric pressure models.
2: Fujita's formula.
3: Mayers' formula.

LINE 5 LEVELOFF LIMIT_WIND_DEP C1 C2 RF_SPUW TYDIR

The flags are the integer.

LEVELOFF the flag activate the limitation of wind speeds in the surge module. This flag only works for only the surge runs.
0: no activate the limitation.

Limit_wind_dep the users assign the wind speed for the limitation.
[limit_wind_dep in m/s] : assign the wind speed for the limitation

C1 the flag is the reduction factor for the wind speed at 10m relating with the typhoon's movement. This flag is working with the Fujita and Mayers' formulas.

C2 the flag is the reduction factor for the wind speed at 10m relating with the gradient wind speed. This flag is working with the Fujita, Mayers, and Mitsuyasu and fujii's formulas.

RF_SPUW	the flag is the reduction factor for the wind speed at 10m for only the Mitsuyasu and fujii's formula. 1: No reduction. [RF_SPUW]: as assigning the value less than 1, the 10m wind speed is reduced.
TYDIR	the user can assign the degree between the gradient wind and the 10m wind speed. [TYDIR in degree] : assign the specific degree. The general value is 30 degree.

LINE 6 OUTFFORM

The flag is the string.

OUTFFORM	the user can assign the format of the output files in the fortran-formatted or unformatted files. As assigning "unformat", the file size will be small. format: activate the ascii file. unformat: activate the unformatted file (binary file).
----------	---

LINE 7 ssl%tab cur%tab met%tab swc%tab ssl%maxout cur%maxout met%maxout

The flags are the integer.

ssl%tab	the flag activates the printout of spatial distributions for the sea surface level. The user can assign the specific time in minutes. 0: no activate it.
cur%tab	the flag activates the printout of spatial distributions for the current. The user can assign the specific time in minutes. 0: no activate it.
met%tab	the flag activates the printout of spatial distributions for the wind and sea level pressure. The user can assign the specific time in minutes.

0: no activate it.

swc%tab the flag activates the printout of spatial distributions for the wave information. The detailed flags for the wave information will be found in the section of "#d.swn".

1: activate the printout

0: no activate it.

ssl%maxout The time interval to calculate the maximum sea surface level

cur%maxout The time interval to calculate the maximum current

met%maxout The time interval to calculate the maximum wind and pressure

LINE 8	rad%tab	psvssb%tab	psvbot%tab	flu%tab	sd%tab	bdc%tab
	upeak_con%tab					

The flags are the integer.

rad%tab the flag activates the printout of spatial distributions for the wave-induced radiation stress. The user can assign the specific time in minutes.

0: no activate it.

psvssb%tab the flag activates the printout of spatial distributions for the sea surface stress. The user can assign the specific time in minutes.

0: no activate it.

psvbot%tab the flag activates the printout of spatial distributions for the bottom stress. The user can assign the specific time in minutes.

0: no activate it.

flu%tab the flag activates the printout of spatial distributions for the flux. The user can assign the specific time in minutes.

In the present version, the flag is not working and will be updated to work it.

0: no activate it.

sd%tab the flag activates the printout of spatial distributions for the sea surface drag coefficient. The user can assign the specific time in minutes.

0: no activate it.

bdc%tab the flag activates the printout of spatial distributions for the bottom drag coefficient. The user can assign the specific time in minutes.

0: no activate it.

upeak_con%tab the flag activate the printout of spatial distributions for the peak current. The specific time in minutes can be assigned.
0: no activate.

LINE 9 ssl%ts cur%ts met%ts

The flags are the integer. The time series is written out as activating the flag "ST" in Line 5 in the file "#d.nest".

ssl%ts the flag activates the printout of time series for the sea surface level. The user can assign the specific time in minutes.
0: no activate it.

cur%ts the flag activates the printout of time series for the current. The user can assign the specific time in minutes.
0: no activate it.

met%ts the flag activates the printout of time series for the wind and sea level pressure. The user can assign the specific time in minutes.
0: no activate it.

LINE 10 rad%ts psvssb%ts psvbot%ts flu%ts sdc%ts bdc%ts

The flags are the integer. The time series is written out as activating the flag "ST" in Line 5 in the file "#d.nest".

rad%ts the flag activates the printout of time series for the wave-induced radiation stress. The user can assign the specific time in minutes.
0: no activate it.

psvssb%ts the flag activates the printout of time series for the sea surface stress. The user can assign the specific time in minutes.
0: no activate it.

psvbot%ts the flag activates the printout of time series for the bottom stress. The user can assign the specific time in minutes.

0: no activate it.

flu%ts the flag activates the printout of time series for the flux. The user can assign the specific time in minutes.

0: no activate it.

sd%ts the flag activates the printout of time series for the sea surface drag coefficient. The user can assign the specific time in minutes. In the present version, the flag is not working and will be updated to work it.

0: no activate it.

bdc%ts the flag activates the printout of time series for the bottom drag coefficient. The user can assign the specific time in minutes.

In the present version, the flag is not working and will be updated to work it.

0: no activate it.

LINE 11 ch manning_n bottomf_con runup_con

The flags are the real.

CH as specifying the sea surface level in meters, the users can rise the mean water level.

MANNING_N the user can assign the desired constant Manning coefficient when BOTTOMF_CON=0 and 1 or

the bottom friction drag coefficient, f_c when BOTTOMF_CON=2, 3 and 4.

2: $f_c/8$

3: $f_c(i,j)/8$

4. $C_d = (0.4 / (1 + \log(0.001/dd)))$

($f_c=0.02$ equal to $C_d=0.0025$)

BOTTOMF_CON the flag can activate the spatially varying manning coefficients. If it is activated, the file of "bottomcoef_01_table.dat" should be needed. See 4.2.5.

0: the constant manning's coefficient designated in "MANNING_N"

- 1: the spatially varying manning's coefficient.
- 2: the friction drag, f_c , is activated.
3. the spatially varying friction drag, f_c
4. the drag coefficient C_d by the quadratic friction law (p.53, Dynamics of marine sands), which will follow the Eq. $C_d = (0.4 / (1 + \log(0.001/dd)))$.

RUNUP_CON the flag activates the inundation calculation in SuWAT as setting "1". If it is activated, the file "*domain_01_runup.table.bin*" is necessary. See, 4.2.4 for the file description.
 0: no activate the inundation calculation.
 1: activate the inundation calculation.

LINE 12 IGR JGR

It is the integer.

IGR the flags indicates the number of the grids along the horizontal direction.
 JGR the flag indicates the number of the grids along the vertical direction.

LINE 13 XOGG YOGG DLSSX DLSSY

The flags are the real. See Fig. 4.1.

XOGG The horizontal length in meters from the origin of the computational domain to the left and lower coner of grid (1,1) in Fig.4.1.
 YOGG The vertical length in meters from the origin of the computational domain to the left and lower coner of grid (1,1) in Fig.4.1.
 DLSSX The grid size along the horizontal direction.
 DLSSY The grid size along the vertical direction.

LINE 14 DLT

The flag is the real.

DLT the flag indicates the time step for the storm surge modules in seconds.
Every domain has to have the identical time steps.

LINE 15 ITLM PRCT

The flags are the integer.

ITLM not use but do not omit.

PRCT not use but do not omit.

LINE 16 IYEAR3 IMON3 IDAY3 IHOURL3 HIMIN3

Format : I4,3I2,F2.0. This date is used to calculate the tide inUTC.

1. The function of the flags is that the proceeding status is shown in the terminal window.
2. In case of with the tide, the flags are the start time in the storm surge modules.
3. In case of without the tide, the flags just shows the proceeding status to the terminal window and do not affect the calculation result.

IYEAR3 year

IMON3 month

IDAY3 day

IHOURL3 hour

Himin3 minute

LINE 17 ITS, ITSS

The flags are the integer.

ITS the flag indicates the time interval in seconds for the information exchange between the modules of surges and waves.

ITSS the flag indicates the time interval in seconds for the information exchange only from the wave module to the surge module. Do not use for the operational run.

LINE 18 WV

The flag is the real.

WV the flag activates the reduction factor for the tide imposed by the tidal prediction model on open boundaries. The default value is "1".
[WV] > 1: the amplification
[WV] < 1 : the reduction
[WV] = 1 : No change

LINE 19 XLN1

The flags are the real. The location is the origin of the computational domain.

XLN1 longitude in degree.

LINE 20 YLT1

The flags are the real. The location is the origin of the computational domain.

YLT1 latitude in degree.

LINE 21 ETOX ETOY ETDX

The flags are the real. See Fig.4.1.

ETOX the longitude in degree of the center of grid in the right and lower corner in the outermost domain.

ETOY the latitude in degree of the center of grid in the right and lower corner in the outermost domain.

ETDX the grid size in degree in the vertical direction.

LINE 22 STOX STOY STDX

The flags are the real. See Fig. 4.1.

STOX the longitude in degree of the center of grid in the left and lower corner in the outermost domain.

STOY the latitude in degree of the center of grid in the left and lower corner at in the outermost domain.

STDX the grid size in degree in the horizontal direction.

LINE 23 WTOX WTOY WTDX

The flags are the real. See Fig. 4.1.

WTOX the longitude in degree of the center of grid in the left and lower corner in the outermost domain.

WTOY the latitude in degree of the center of grid in the left and lower corner in the outermost domain.

WTDX the grid size in degree in the vertical direction.

LINE 24 NTOX NTOY NTDX

The flags are the real. See Fig. 4.1.

NTOX the longitude in degree of the center of grid in the left and upper corner in the outermost domain.

NTOY the latitude in degree of the center of grid in the left and upper corner in the outermost domain.

NTDX the grid size in degree in the horizontal direction.

LINE 25 debugging

debugging 1: activate the printout of some values for one grid.
 0: non activate debugging

LINE 26 idebug jdebug ktdebug

The present version support only one grid to print out some values.

idebug	the grid number in the horizontal direction
jdebug	the grid number in the vertical direction
ktdebug	the time step to stop the run

5.5 #d.nest

- The nested domain information is given in the files.
- The grid IDs are assigned to write out the time series at the position desired in Line 6 in the file "#d.nest"

#d.nest

Line number	Flags			
Line 1	ID	DTO	DTN	
Line 2	TXX	TYT		
Line 3	I_FINE	J_FINE	DISX	DISY
Line 4	M1	M2	M3	M4
Line 5	ST	IDN		
Line 6	pointx	pointy	comment (option)	

LINE 1 ID DTO DTN

The flags are the integer.

ID not use but does not change and keep it.

DTO not use but does not change and keep it.

DTN not use but does not change and keep it.

LINE 2 TXX TYT

The flags are the integer.

TXX	the flag indicates the number of the grids along the horizontal direction in the nested domain. Same with IGR in the file of "02.con".
TTY	the flag indicates the number of the grids along the vertical direction in the nested domain. Same with JGR in the file of "02.con".

LINE 3 I_FINE J_FINE DISX DISY

The flags are the real.

I_FINE	the flag indicates the horizontal length from the origin of the computational domain to the grid (1,1) in the nested domain. Same with XOGG in the file of "02.con".
J_FINE	the flag indicates the vertical length from the origin of the computational domain to the grid (1,1) in the nested domain. Same with YOGG in the file of "02.con".
DISX	the flag indicates the grid size in meters in the horizontal direction of the nested domain. Same with DLSSX in the file of "02.con".
DISY	the flag indicates the grid size in meters in the vertical direction of the nested domain. Same with DLSSY in the file of "02.con".

LINE 4 M1 M2 M3 M4

The flags are the integer.

M1	not use but does not change and keep it.
M2	not use but does not change and keep it.
M3	not use but does not change and keep it.
M4	not use but does not change and keep it.

LINE 5 ST IDN

The flags are the integer.

ST the flag indicates the number of stations for the printout. The sea surface level, the current, the sea level pressure and the wind are written out with the time interval of the flags designated in Line 8 and Line 10 of the file "#d.con".

IDN not use but keep it as 3.

LINE 6 POINTX POINTY

The flags are the integer. The number of the stations should be same with the number of "ST" in Line 5.

POINTX the flag indicates the grid ID in the horizontal direction the user wants.

POINTY the flag indicates the grid ID in the vertical direction the user wants.

5.6 swaninit#

The default input file for the wave module of SWAN.

4	
DPRI at Kyoto University	
3	
0#.swn	
4	
	test file name
6	
99	
\$	comment identifier
/	dir sep char in input file
/	dir sep char replacing previous one
1	default time coding opti

5.7 #d.SWN

PROJECT 'Numer' '01'

!##### MODEL INPUT #####

SET level nor depmin maxmes maxerr grav rho inrhog hsrerr CARTesian pwtail froud-
max printf prtest

MODE DYNAMIC TWODimensional

COORDinates CARTesian

CGRID REGular xpc ypc alpc xlenc ylenc mxc myc CIRcle mdc flow fhigh msc

INP BOTtom REG xpinp **ypinp** 0 mxinp myinp dxinp dyinp EXCEPTION

INP WLEVel REG xpinp **ypinp** 0 mxinp myinp dxinp dyinp NONSTAT tbeginp deltinp Sec
tendingp

INP VX REG xpinp = XOGG + DLSSX ypinp 0 mxinp myinp dxinp dyinp NONSTAT tbeginp
deltinp Sec tendingp

INP VY REG xpinp ypinp=YOGG+DLSSY 0 mxinp myinp dxinp dyinp NONSTAT tbeginp deltinp
Sec tendingp

INP WIND REG xpinp = XOGG + DLSSX **ypinp=YOGG+DLSSY** 0 mxinp myinp dxinp dyinp
NONSTAT tbeginp deltinp Sec tendingp

READinp BOTtom 1 fname 1 0 FREE

READinp WLevel 1 SERIES fname 1 0 FREE

READinp CUREnt 1 SERIES fname 1 0 FREE

READinp WIND 1 SERIES fname 1 0 FREE

BOUND SHAPespec JONswap 3.3 PEAK DSPR POWER

!BOUNDnest1 NEST 'd01.out.nest.swt' CLOSeD|OPEN

INITial ZERO

GEN3 JANSSEN 4.5 0.5 AGROW 0.0015

QUADrupl iquad=2 lambda=0.25 cnl4=30000000. csh1=5.5 csh2=0.8571 csh3=-1.25

BREAKING CONSTANT alpha=1.0 gamma=0.73

FRICTION JONSWAP cfjon=0.067

TRIAD trfac=0.1 cutfr=2.2 urcrit=0.2 urslim=0.01

LIMiter ursell=10.0 qb=1.0

SETUP supcor=0.

DIFFRac idiffr=1 smpar=0. smnum=0. cgmod=1.

PROP BSBT

!##### OUTPUT REQUESTS #####

GROUP 'GRID2' SUBGRID ix1= 0 ix2= mxinp iy1= 0 iy2= myinp
NGRID 'NEST' xpn ypn alpn xlenn ylenn mxn myn

POINTS 'p1' lengthx1 lengthy1

OUTPut OPTIons '%' TABle field=12
OUTPut OPTIons '%' BLOck ndec=9 len=6
OUTPut OPTIons '%' SPEC ndec=9

BLOCK 'GRID2' HEADER 'rs_d0#.swt' LAY-OUT IDLA=1 FORCE OUTput tbegblk deltblk
SEC

BLOCK 'GRID2' HEADER 'max_hsign_d0#.swt' LAY-OUT IDLA=1 HSIGN OUTput tbegblk deltblk
SEC

BLOCK 'GRID2' HEADER 'max_direction_d0#.swt' LAY-OUT IDLA=1 PDIR OUTput tbegblk deltblk
SEC

BLOCK 'GRID2' HEADER 'max_period_d0#.swt' LAY-OUT IDLA=1 RTP OUTput tbegblk deltblk
SEC

TABLE 'p1' HEAD 'p1_d01.swt' TIME XP YP HSWE WLEN HS PDIR RTP PER DEP VEL WIND
FORCE OUT tbegblk deltblk SEC

NESTOUT 'NEST' 'd01.out.nest.swt' OUTPUT tbegnst deltnst Sec

COMPUte NONSTAT tbegc deltc Sec tendc

STOP

In this manual, the coordinate system in SWAN will be described, which corresponds to the storm surge module described before.

See the user manual on SWAN for Commands for physics in detail.

The coordinate system of the Cartesian in SWAN is same with that in the storm surge module. It is strongly recommended that the format and the order of commands must be kept as long as possible.

The files of "#d.swn", which "#" indicates the domain number, can be written out by running the scripts of "controfile4swan_3D.sh", which is supplied, after making all input files of "01.con", "event_name.srt" and "case.event_name.srt".

5.7.1 The command for model input

- CGRID :
 1. $xpc = XOGG + DLSSX / 2$ in #d.con
 2. $ypc = YOGG + DLSSY / 2$ in #d.con
- INP BOTTOM :
 1. $xpinp = xpc$
 2. $ypinp = ypc$
 3. EXCEPTION -99 is recommended for the complex topography to save the computational time (The neighborhood of the shoreline uses the topography without any modification).
- INP WLEVEL :
 1. The period between *tbeginp* and *tendinp* must be same with the period specified on the storm surge module after the spinup duration.
 2. It is strongly recommended that *deltinp* indicates the time interval of information exchange (= ITS in 01.con).
- INP VX :
 1. $xpinp = XOGG + DLSSX$ in #d.con
 2. $ypinp = ypc$
- INP VY :
 1. $xpinp = xpc$
 2. $ypinp = YOGG + DLSSY$ in #d.con
- INP WIND :
 1. $xpinp = XOGG$ in #d.con
 2. $ypinp = YOGG$ in #d.con

- READinp BOTTOM:
 1. 'fname' = 'domain_01.table.geo.swan'. See 4.3 for the detail.
- READinp WLEVEL :
 1. 'fname' = 'seri.wle'. Do not change and keep it in the "#d.swn". See 4.3 for the detail.
- READinp CURRENT :
 1. 'fname' = 'seri.cur'. Do not change and keep it in the #d.swn. See 4.3 for the detail.
- READinp WIND :
 1. 'fname' = 'seri.win' - Do not change and keep it in the #d.swn. See 4.3 for the detail.
- BOUNDSPEC : The command is used in only the 1d.swn or remove it to use the default setting of SWAN.
 1. EAST : Specify the open boundary in the east in 1d.swn
 2. NORTH : Specify the open boundary in the north in 1d.swn
 3. SOUTH : Specify the open boundary in the south in 1d.swn
 4. WEST : Specify the open boundary in the west in 1d.swn
- BOUNDNEST1 : The command is used in the nested domains from 02.swn instead of the command of "BOUNDSPEC".
 1. 'd01.out.nest.swt': The file includes the wave information for the open boundaries in the nested domains and the path should be indicated in the file "filename_d0#.con".

5.7.2 The commands for output requests

- GROUP : the domain information in the present domain.
 1. ix1 = 0
 2. ix2 = mxinp, the number of grids in the horizontal direction.
 3. iy1 = 0
 4. iy2 = myinp, the number of grids in the vertical direction.
- NGRID : the nested domain information embedded in the present domain.
 1. xpn = the horizontal length of the nested domain from the origin of the whole domain.
 2. ypn = the vertical length of the nested domain from the origin of the whole domain.
 3. xlenn = the horizontal length from (1,1) to (imax,1) in the nested domain.
 4. ylenn = the horizontal length from (1,1) to (1,jmax) in the nested domain.
 5. mxn = the number of grids in the horizontal direction for the nested domain.

6. myn = the number of grids in the vertical direction for the nested domain.

➤ BLOCK :

1. FORCE : wave-induced radiation stress, deltblk should be same with the time interval of the information exchange (=ITS in d#.con and = deltic in d#.swn).
2. HSIGN : significant wave height, deltblk is optionally selected by users want.
3. PDIR : wave direction, deltblk is optionally selected by users want.
4. RTP : wave period, deltblk is optionally selected by users want.

➤ NESTOUT ;

1. NEST : the wave information along the boundaries of the nested domains , deltnst is same with the time interval of the information exchange (=ITS in d#.con and = deltic in d#.swn)

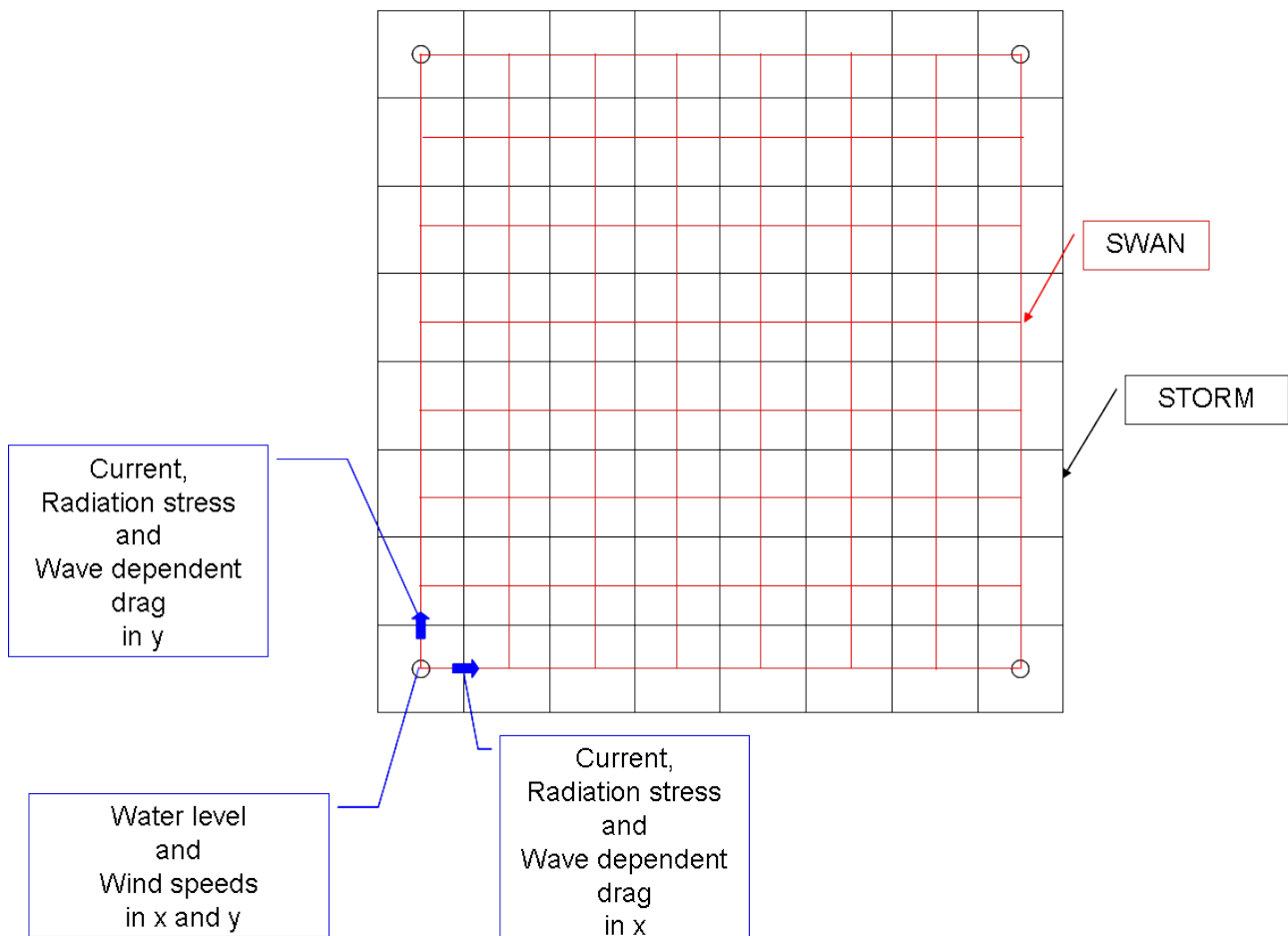


Fig. 4.4 Scheme of the computational grid in the wave module with the storm surge module.

6. Output files for the storm surge module

- The formatted file has the extension "swt".
- The unformatted file has the extension "swt.un".
- "#" indicates the domain number.
- The files of the maximum and minimum values are automatically written out as activating the flags.

6.1 d0#.out.nest.swt

The wave spectra information specified in the command of NESTOUT in "0#.swn" is temporally saved in this file.

6.2 Spatial distributions I

The spatial distributions of the sea surface level, the current, the wind and the sea level pressure in the domain assigned by the time interval of *ssl%tab*, *cur%tab*, *met%tab* in Line 7 in the file "#d.con".

Flag	Output file name	Variable
ssl%tab	ssl_d0#.swt	sea surface level
	max_wse_d0#.swt	maximum values
	min_wse_d0#.swt	minmum values
cur%tab	cur_x_d0#.swt	current along the horizontal direction
	cur_y_d0#.swt	current along the vertical direction
	max_crv_in_x_d0#.swt	maximum values
	min_crv_in_x_d0#.swt	minmum values
	max_crv_in_x_d0#.swt	maximum values
	min_crv_in_x_d0#.swt	minmum values
met%tab	u10_d0#.swt	wind speed along the horizontal direction
	v10_d0#.swt	wind speed along the vertical direction

	slp_d0#.swt	sea level pressure
--	-------------	--------------------

6.3 Time series I

Time series of the sea surface level, the current, the wind and the sea level pressure in the domain assigned by the time interval of *ssl%ts*, *cur%ts* and *met%ts* in Line 9 in the file "*#d.con*".

Flag	Output file name	Variable
ssl%ts	timeseriesssl_d0#.swt	sea surface level
cur%ts	timeseries_crv_in_x_d0#.swt	current along the horizontal direction
	timeseries_crv_in_y_d0#.swt	current along the vertical direction
met%ts	timeseriesu10_d0#.swt	wind speed along the horizontal direction
	timeseriesv10_d0#.swt	wind speed along the vertical direction
	timeseriesslp_d0#.swt	sea level pressure

6.4 Spatial distributions II

The spatial distributions in the domain assigned by the time interval of *rad%tab*, *psvssb%tab*, *psvbot%tab*, *flu%tab*, *sdv%tab* and *bdc%tab* in Line 8 in the file "*#d.con*".

Flag	Output file name	Variable	Note
rad%tab	rads_x_d0#.swt	wave-induced radiation stress along the horizontal direction	
	rads_y_d0#.swt	wave-induced radiation stress along the vertical direction	
	max_rad_in_x_d0#.swt	maximum values	
	min_rad_in_x_d0#.swt	minmum values	
	max_rad_in_x_d0#.swt	maximum values	
	min_rad_in_x_d0#.swt	minmum values	

psvssb%tab	sfs_x_d0#.swt	sea surface stress along the horizontal direction	
	sfs_y_d0#.swt	sea surface stress along the vertical direction	
	max_sfs_in_x_d0#.swt	maximum values along the horizontal direction	
	min_sfs_in_x_d0#.swt	minmum values along the horizontal direction	
	max_sfs_in_x_d0#.swt	maximum values along the vertical direction	
	min_sfs_in_x_d0#.swt	minmum values along the vertical direction	
psvbot%tab	bts_x_d0#.swt	bottom stress along the horizontal direction	
	bts_y_d0#.swt	bottom stress along the vertical direction	
	max_bts_in_x_d0#.swt	maximum values along the horizontal direction	
	min_bts_in_x_d0#.swt	minmum values along the horizontal direction	
	max_bts_in_x_d0#.swt	maximum values along the vertical direction	
	min_bts_in_x_d0#.swt	minmum values along the vertical direction	
flu%tab	flu_x_d0#.swt	flux along the horizontal direction	In the present version, the flag is not working and will be updated soon.
	flu_y_d0#.swt	flux along the vertical direction	In the present version, the flag is not working and will be updated soon.
	max_flx_in_x_d0#.swt	maximum values along the horizontal direction	
	min_flx_in_x_d0#.swt	minmum values along the horizontal direction	
	max_flx_in_x_d0#.swt	maximum values along the vertical direction	
	min_flx_in_x_d0#.swt	minmum values along the vertical direction	

sd%tab	xcd_d0#.swt	sea surface drag coefficient in the horizontal direction	
	ycd_d0#.swt	sea surface drag coefficient in the vertical direction	
	max_cd_d0#.swt	maximum values	
	min_cd_d0#.swt	minmum values	
bdc%tab	friction_velocity_d0#.swt	friction velocity near the sea surface	
	aerodynamic_roughness_zo_d0#.swt	aero dynamic roughness (Zo)	
	aerodynamic_roughness_ze_d0#.swt	aero dynamic roughness (Ze)	
	tauw_d0#.swt	wave-induced stress near the sea surface	
	tautout_d0#.swt	total stress near the sea surface	

6.5 Time series II

Time series in the domain assigned by the time interval of *rad%tab*, *psvssb%tab*, *psvbot%tab*, *flu%tab*, *sd%tab* and *bdc%tab* in Line 10 in the file "#d.con".

Flag	Output file name	Variable	Note
rad%ts	timeseries_rad_in_x_d0#.swt	wave-induced radiation stress along the horizontal direction	
	timeseries_rad_in_y_d0#.swt	wave-induced radiation stress along the vertical direction	
psvssb%ts	timeseries_sfs_in_x_d0#.swt	sea surface stress along the horizontal direction	
	timeseries_sfs_in_y_d0#.swt	sea surface stress along the vertical direction	
psvbot%ts	timeseries_bts_in_x_d0#.swt	bottom stress along the horizontal direction	

	timeseries_bts_in_y_d0#.swt	bottom stress along the vertical direction	
flu%ts	timeseries_flu_in_x_d0#.swt	flux along the horizontal direction	
	timeseries_flu_in_y_d0#.swt	flux along the vertical direction	
sd%ts	xcd_d0#.swt	sea surface drag coefficient in the horizontal direction	In the present version, the flag is not working and will be updated soon.
	ycd_d0#.swt	sea surface drag coefficient in the vertical direction	In the present version, the flag is not working and will be updated soon.
bdc%ts	friction_velocity_d0#.swt	friction velocity near the sea surface	In the present version, the flag is not working and will be updated soon.
	aerodynamic_roughness_zo_d0#.swt	aero dynamic roughness (Zo)	In the present version, the flag is not working and will be updated soon.
	aerodynamic_roughness_ze_d0#.swt	aero dynamic roughness (Ze)	In the present version, the flag is not working and will be updated soon.
	tauw_d0#.swt	wave-induced stress near the sea surface	In the present version, the flag is not working and will be updated soon.
	tautout_d0#.swt	total stress near the sea surface	In the present version, the flag is not working and will be updated soon.

6.6 Summary of "event_name.srt"

The files of "ty_data_d0#.list.swt" are written out on every domain.

Contents:

Hourly No	Longitude	Latitude	depression rate of central pressure	Maximum radius	Moving speed of typhoon along the horizontal direction	Moving speed of typhoon along the vertical direction

7. Output files for the wave module

Except **FORCE**, other wave parameters are optional. If the commands of BLOCK in #d.SWN is activated, the following files are written out. The rest of wave parameters can be arranged after the command of RTP.

7.1 **FORCE**: wave-induced radiation stresses. No option.

Flag	Output file name	Variable
FORCE	max_rs_x_d0#.swt	the maximum values of radiation stress along the horizontal direction
	max_rs_y_d0#.swt	the maximum values of radiation stress along the vertical direction
	rad_x_d0#.swt	the spatial distribution of radiation stress along the horizontal direction
	rad_y_d0#.swt	the spatial distribution of radiation stress along the vertical direction

7.2 HSIGN: significant wave height (option)

Flag	Output file name	Variable
HSIGN	max_rs_x_d0#.swt	the maximum values of radiation stress along the horizontal direction
	max_rs_y_d0#.swt	the maximum values of radiation stress along the vertical direction
	rad_x_d0#.swt	the spatial distribution of radiation stress along the horizontal direction
	rad_y_d0#.swt	the spatial distribution of radiation stress along the vertical direction

7.3 PDIR: direction (option)

Flag	Output file name	Variable
PDIR	max_direction_d0#.swt	the maximum values of wave directions
	direction_d0#.swt	the spatial distribution of wave directions

7.4 RTP: period (option)

Flag	Output file name	Variable
RTP	max_period_d0#.s wt	the maximum values of wave periods
	period_d0#.swt	the spatial distribution of wave periods

8. Frequent Q&A

8.1 Garden-Sprinkler-Effect (GSE)

To solve this problem particularly on coarse domains that have large grid sizes larger than 1 km, it is recommended that remove one/two grid island and change to flat water depths with neighbouring water depths.